PCBC H Parametric & Sequences Test (Practice C) **Name**_____ _____/27

SHOW ALL WORK FOR FULL CREDIT. ***Calculator*** **You have the entire period.**

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

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_____ _____

Signature

Solve the following application problem. Report answers to the thousandths. Include units in your answers. (4 points)

7. A rock is thrown from a 20-foot cliff at an angle of 15° with an initial speed of 45 ft/s.

a) Write a set of parametric equations for the motion of the rock. (Recall:
$x = (v\cos\theta)t,\ \ y = (v\sin\theta)t - 16t^2 + h$)

b) Determine the maximum height reached by the rock.

c) Determine how long the rock is in the air.

d) Determine how far the rock traveled in the air.

The given pattern continues. Write down the nth term of the sequence $\{a_n\}$ suggested by the pattern. (3 points)

8. $3, -\frac{5}{4}, \frac{7}{9}, -\frac{9}{16}, \ldots$

Write out the sum. (3 points)

9. $\sum_{k=0}^{n} \frac{(-2)^k}{k+1}$

Find each sum. Show the properties and formulas you use, then use your calculator. (#10 and #11 are 3 points each, #12 is 4 points)

10. $\sum_{k=1}^{40} (3k+4)$

11. $\sum_{k=1}^{9} \left(k^3+2k\right)$

12. $\sum_{k=12}^{30} \left(4k^2 - 3k\right)$

Expand using the Binomial Theorem. (#13 is 3 points, #14 is 4 points)

13. $(x+1)^7$

14. $(2x+3)^4$

Extra practice: more sequence and summation questions

Not part of the 27 points. Worked in the key.

Express the sum using summation notation.

E1. $4-8+16-32+\cdots+1024$

The given pattern continues. Write down the nth term of the sequence $\{a_n\}$ suggested by the pattern.

E2. $1,\ \frac{1}{8},\ \frac{1}{27},\ \frac{1}{64},\ \ldots$

Answer Key

Work on the left, the reason beside it. Boxed = final answer.

Day 2: Calculator

7. Rock: $h = 20$ ft, $\theta = 15°$, $v = 45$ ft/s. Calculator in **degree** mode, Graph in **Parametric**.

a) $x = (45\cos 15°)t$ — *plug v and θ into the given formula; degree symbol required*

$y = (45\sin 15°)t - 16t^2 + 20$ — *+h: starts at 20 ft*

$x_1(t) = 45\cos(15°)\cdot t,\ \ y_1(t) = 45\sin(15°)\cdot t - 16t^2 + 20$ — *what you type in*

$0 \le t \le 2$, tstep 0.01; x : 0 to 80, y : 0 to 28 — *graph doesn't come back down? raise tmax*

b) $\boxed{y \approx 22.120 \text{ ft}}$ — *trace to the top ($t \approx 0.364$); not Analyze, Maximum*

c) $\boxed{t \approx 1.540 \text{ s}}$ — *trace to $y \approx 0$; trace step 0.01, then 0.001*

d) $\boxed{x \approx 66.928 \text{ ft}}$ — *same point, read x*

Units: feet (the speed was ft/s).

8. $\frac{1}{3}, -\frac{2}{5}, \frac{3}{7}, -\frac{4}{9}, \ldots$

top: $2n+1$, bottom: n^2 — *write 1 2 3 4 above; top and bottom separately*

$+,-,+,- \Rightarrow (-1)^{n+1}$ — *first term positive*

$\boxed{a_n = (-1)^{n+1}\frac{2n+1}{n^2}}$ — *write 3 as $\frac{3}{1}$ to see the pattern*

9. $\sum_{k=0}^{n} \frac{(-2)^k}{k+1}$

$\frac{(-2)^0}{1} + \frac{(-2)^1}{2} + \frac{(-2)^2}{3} + \cdots + \frac{(-2)^n}{n+1}$ — *start at $k = 0$*

$\boxed{1 - 1 + \frac{4}{3} - \cdots + \frac{(-2)^n}{n+1}}$ — *don't add*

10.

$3\sum_{k=1}^{40} k + 4\sum_{k=1}^{40} 1$ — *(2), (3) split; (1) constants out*

$3\left[\frac{40(41)}{2}\right] + 4(40)$ — *(6), (5) with $n = 40$; then the calculator*

$\boxed{2620}$

11.

$\sum_{k=1}^{9} k^3 + 2\sum_{k=1}^{9} k$ — *(2), (3) split; (1) constants out*

$\left[\left[\frac{9(10)}{2}\right]^2\right] + 2\left[\frac{9(10)}{2}\right]$ — *(8), (6) with $n = 9$; then the calculator*

$\boxed{2115}$

12.

$4\sum_{k=12}^{30} k^2 - 3\sum_{k=12}^{30} k$ — *(2), (3) split; (1) constants out*

$4\left[\sum_{k=1}^{30} k^2 - \sum_{k=1}^{11} k^2\right] - 3\left[\sum_{k=1}^{30} k - \sum_{k=1}^{11} k\right]$ — *(4): starts at 12, so $j = 11$*

$4\left[\frac{30(31)(61)}{6} - \frac{11(12)(23)}{6}\right] - 3\left[\frac{30(31)}{2} - \frac{11(12)}{2}\right]$ — *(7), (6); then the calculator*

$\boxed{34599}$

13. $(x+1)^7$: $n = 7$, row 1 7 21 35 35 21 7 1. The sign goes with its term.

$1x^7(1)^0 + 7x^6(1)^1 + 21x^5(1)^2 + 35x^4(1)^3$ — *blanks first: first term's powers count down, second term's count up*

$+35x^3(1)^4 + 21x^2(1)^5 + 7x^1(1)^6 + 1x^0(1)^7$

$\boxed{x^7 + 7x^6 + 21x^5 + 35x^4 + 35x^3 + 21x^2 + 7x + 1}$

Second term positive: every term positive.

14. $(2x+3)^4$: $n = 4$, row 1 4 6 4 1. The sign goes with its term.

$1(2x)^4(3)^0 + 4(2x)^3(3)^1 + 6(2x)^2(3)^2 + 4(2x)^1(3)^3$ — *blanks first: first term's powers count down, second term's count up*

$+1(2x)^0(3)^4$

$(2x)^4 = 16x^4, \ldots$ — *the whole first term gets the power*

$\boxed{16x^4 + 96x^3 + 216x^2 + 216x + 81}$

Second term positive: every term positive.

Extra practice (not scored)

E1. $4 - 8 + 16 - \cdots + 1024$

$2^2, -2^3, 2^4, \ldots, 2^{10}$ — *powers of 2 from 2^2; $1024 = 2^{10}$*

even powers $+ \Rightarrow (-1)^k$

$\boxed{\sum_{k=2}^{10} (-1)^k 2^k}$ — *$\sum_{k=2}^{10}(-2)^k$ is the same*

E2. $\frac{1}{1}, \frac{2}{8}, \frac{3}{27}, \frac{4}{64}, \ldots$

top: 1, bottom: n^3 — *write 1 2 3 4 above; top and bottom separately*

$\boxed{a_n = \frac{1}{n^3}}$ — *cubes*